

Properties of the MLE: Uncertainty in Parameters

INFO 521 · student notes · Module 3 · Lecture B · ~7 min read

Lecture A produced $\hat{\mathbf{w}}$. This lecture puts it on trial with three questions: is it unique, is it unbiased, and how uncertain is it?

Uniqueness

The gradient of the log-likelihood vanishes at the normal equations, and its second derivative — the **Hessian** — comes out constant, $\mathbf{H} = -\frac{1}{\sigma^2} \mathbf{X}^\top \mathbf{X}$. This is **negative definite**, so the log-likelihood is strictly concave: one hump, no ridges, a single global maximum. It is the Module 1 uniqueness argument again — the same full-column-rank $\mathbf{X}^\top \mathbf{X}$, now wearing the name “negative-definite Hessian.”

The generative picture and unbiasedness

Uncertainty is defined by imagining **repeated datasets**: run the generating process again and again, fit each one, and the estimates scatter into a cloud. The spread of that cloud is the covariance of $\hat{\mathbf{w}}$.

Taking the expectation of $\hat{\mathbf{w}}$ over that process — with $\mathbf{X}$ fixed and $\mathbb{E}[\mathbf{y}] = \mathbf{X}\mathbf{w}$ — the inverse cancels its own matrix and leaves $\mathbb{E}[\hat{\mathbf{w}}] = \mathbf{w}$: the estimator is **unbiased**. Be careful what that means. It is a statement about the *center* of the cloud across many size- N samples. It is *not* the claim that more data helps (a separate sample-size effect), *not* the stronger minimum-variance (UMVU) property, and *not* a promise that $\hat{\mathbf{w}}$ is best for your one dataset.

Covariance and Fisher information

The spread of the cloud has a closed form,

$$\text{Cov}[\hat{\mathbf{w}}] = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}.$$

Read it for intuition: low noise (small σ^2) and abundant, well-spread data (large $\mathbf{X}^\top \mathbf{X}$) both sharpen the estimate. The full derivation is in the [covariance appendix](#).

The same quantity has a deeper name. The **Fisher information** is the expected curvature of the log-likelihood, $\mathcal{J} = \frac{1}{\sigma^2} \mathbf{X}^\top \mathbf{X}$, so $\text{Cov}[\hat{\mathbf{w}}] = \mathcal{J}^{-1}$. Curvature and uncertainty are inverses: a sharply curved peak pins the parameters tightly; a flat one leaves them loose. On NHANES, a covariance ellipse from 40 patients is wide and fuzzy while the full-sample ellipse collapses to a tight blob — the payoff of data, drawn.

The demo

The parameter-uncertainty explorer lets you **Draw one dataset** (a single $\hat{\mathbf{w}}$ dot), **Draw 100** (the cloud floods the theoretical 2σ ellipse, centred on the true $\mathbf{w}$ — unbiasedness made visible), and nudge the noise or sample-size sliders to watch the ellipse grow or shrink.

What to remember

- The MLE is unique (concave ℓ , negative-definite Hessian), unbiased ($\mathbb{E}[\hat{\mathbf{w}}] = \mathbf{w}$), and quantifiably uncertain.
- $\text{Cov}[\hat{\mathbf{w}}] = \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1} = \mathcal{I}^{-1}$ — the inverse Fisher information.
- Unbiasedness is about the center of the estimate-cloud, not any single fit.
- Checkpoint quiz 1.2 (Week 3, D2L) gates milestone 1.2 on the uniqueness and unbiasedness arguments; the MLE == OLS numerical check is in Homework Unit 3.

Through-lines

One matrix does three jobs across modules: the negative-definite Hessian (uniqueness) is, in expectation, the Fisher information, whose inverse becomes the **Laplace posterior covariance** in Module 5. Keep your eye on $\mathbf{X}^\top \mathbf{X}$.